

BIOSENSING PLATFORM & ML CLASSIFICATION SYSTEM
Technical Documentation
Piezo Pico-to-Femtotechnology Sensors Inc.
Version 1.0 — August 2026

1. Overview

This documentation presents the API architecture supporting the patented Piezo Pico-to-Femtotechnology Sensors Inc. biosensing device, the VCE-100 machine-learning classification models (**V2, V6, and Spectral**), the virus database, and the educational workflow designed for student laboratories. It integrates simulation, classification, noise analysis, virus-range filtering, and model comparison to demonstrate how biosensing data are processed and interpreted.

2. Device Documentation (Patented Technology)

2.1 Device Description

The patented biosensing device detects frequency noise variations down to 10^{-10} Hz and is trained to identify viral mass at resonant or base frequencies. When a virus binds to the sensor surface, the resonant frequency shifts — this change is converted into biosensing features that feed into trained ML models for classification.

Patented Innovations

- Nano-resonant biosensor
- Frequency-shift detection algorithm
- Mass-estimation pipeline
- Multi-class probability classification
- Real-time simulation mode

Device Architecture

- Sensor Module: AT-Cut QCM mass sensor
- Signal Processor: Converts analog resonance into digital features
- ML Engine: Implements V2, V6, and Spectral models
- API Gateway: FastAPI backend
- Dashboard: React-based frontend interface

Device Images

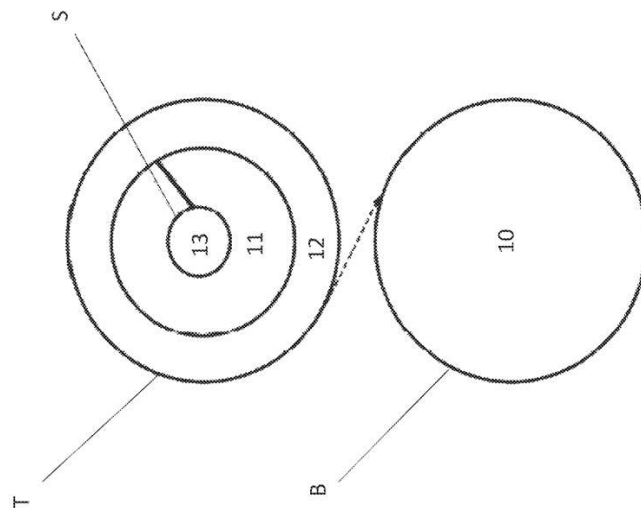


FIG. 1

S = Center dot electrode, T = Quartz disk between top and bottom electrodes, B = Bottom electrode, 10 = Full gold electrode at bottom, 11 = Blank quartz ring, 12 = Gold electrode ring, 13 = Dot center electrode.

3. API Documentation (Swagger & Folder Structure)

The API backend is built with FastAPI, exposing endpoints for simulation and classification. Detailed Swagger specifications and folder structure are documented separately in the following reference files:

- Swagger Documentation: *Swagger.pdf*
- Folder Structure of Selected ML Models:
Folder Structure of Selected ML Models.pdf

3.1 API Architecture

The backend exposes:

- /Simulate — Generates biosensing features
- /Classify/v2 — XBoost model
- /Classify/v6 — RandomForest model
- /Classify/spectral — Spectral signature model

Swagger UI: <http://127.0.0.1:8000/docs> OpenAPI JSON:
<http://127.0.0.1:8000/openapi.json>

4. Virus Table Documentation

The ML models use a 100-virus database containing:

- Virus ID
- Virus Name
- Mass (fg)
- Transmission Mode
- ML Probability

Virus masses were derived from literature and normalized to sensor response. Access to the table is available after subscription to the ML models for one month.

5. ML Model Documentation

5.1 V2 Model — XBoost (12 000 samples)

Training Summary

The V2 classification model was trained using XBoost on 12 000 samples, achieving 88.5 % accuracy.

ML Training V2

Execute

Model Name: Analyzer V2

Dataset Size: Small Dataset (12,000 samples)

Status: Training Completed

Accuracy: 88.5%

Loss: 0.31

Last Trained: 2026-06-30

Epochs: 30

Training Time (min): 9

Logs

Epoch 1/30 - loss: 1.12 - acc: 52%

Epoch 10/30 - loss: 0.61 - acc: 71%

Epoch 20/30 - loss: 0.41 - acc: 81%

Epoch 30/30 - loss: 0.31 - acc: 88.5%

Educational Purpose

Students learn:

- Early-stage ML model behavior
- Noise sensitivity
- Why XBoost cannot identify every virus in the dataset
- How limited datasets affect accuracy

5.2 V6 Model — RandomForest (120 000 samples)

Training Summary

The V6 model was trained using RandomForest on 120 000 samples, achieving 94.2 % accuracy.

ML Training V6

Execute

Model Name: Analyzer V6

Dataset Size: Large Dataset (120,000 samples)

Status: Training Completed

Accuracy: 94.2%

Loss: 0.12

Last Trained: 2026-06-30

Epochs: 50

Training Time (min): 18

Logs

Epoch 1/50 - loss: 0.89 - acc: 62%

Epoch 10/50 - loss: 0.45 - acc: 78%

Epoch 25/50 - loss: 0.22 - acc: 88%

Epoch 50/50 - loss: 0.12 - acc: 94.2%

Educational Purpose

Students learn:

- Ensemble learning principles
- How RandomForest identifies every virus in the trained algorithm
- How larger datasets improve accuracy
- How noise affects classification stability

6. Noise-Based Virus Prediction Tables

Tables 1 and 2(a) illustrate how students can predict viruses at different noise levels.

V2 (XBoost) Noise Table

Students use this table to study how noise influences early-stage ML models.

Table 1: Top Prediction of Virus with VCE-100 v2 (XBoost) at Different Frequency Noises

Noise (Hz)	Virus ID	Mass (fg)	Virus Name	Mode of Transmissions
0.1				
0.01				
0.001				
0.0001				
0.00001				
0.000001				
0.0000001				
0.00000001				
0.000000001				
0.0000000001				
0.1234567890				
0.0123567890				
0.0023672980				
0.0004325678				
0.0000236715				
0.0000072345				
0.0000003567				
0.0000000563				
0.0000000053				

By using table 1 above the students will be able to predict the detected virus at different frequency noises using v2 ML model.

V6 (RandomForest) Noise Table

Students observe that RandomForest remains stable and can identify viruses even at extremely low noise levels.

Table 2 (a): Top Prediction of Virus with VCE-100 v6 (Randomforest) at Different Frequency Noises

Noise (Hz)	Virus ID	Mass (fg)	Virus Name	Mode of Transmissions
0.1				
0.01				
0.001				
0.0001				
0.00001				
0.000001				
0.0000001				
0.00000001				
0.000000001				
0.0000000001				
0.1234567890				
0.0123567890				
0.0023672980				
0.0004325678				
0.0000236715				
0.0000072345				
0.0000003567				
0.0000000563				
0.00000000053				

By using table 2 (a) students will be able to understand that RandomForest uses neural network rather than classification itself to identify every virus included in the model, using XBoost, it is not possible to pinpoint every virus in the trained algorithm.

7. Virus-Range Filtering (Table 2 b)

By using Table 2(b), students can filter viruses with the lowest probabilities to identify or predict the detected virus.

This teaches:

- Probability-based filtering
- Range-based virus narrowing
- How ML models behave across subsets of virus IDs

Table 2 (b): Top Prediction of Virus with VCE-100 v6 (Random Forest) at Different Virus ID Range

Noise (Hz)	Virus ID Range	Mass (fg)	Virus Name	Mode of Transmissions
0.1	1-100			
0.01	1-100			
0.001	1-100			
0.0001	1-100			
0.00001	1-100			
0.000001	1-100			
0.0000001	1-100			
0.00000001	1-100			
0.000000001	1-100			
0.0000000001	1-100			
0.00001	5-100			
0.00001	11-100			
0.00001	59-67			
0.000001	1-100			
0.000001	7-34			
0.000001	40-76			
0.000001	58-100			
0.000001	58-78			
0.000001	79-100			

By using table 2b, the student will be able to filter the virus with least probabilities and identify or predict the detected virus.

8. Sensor Frequencies and Noise

Side-by-Side V2 & V6 Comparison

Threshold (Hz): 0.00001, Virus ID Range: 1 – 100

Sensor Frequencies

- Base Frequency (Hz): **1 694 137.606 239 511 9**
- Measured Frequency (Hz): **1 694 138.316 002 553**

Students adjust frequency noise and observe how V2 and V6 predict viruses as a function of mass and virus ID as seen in the display panel below.

Side-by-Side V2 & V6 Comparison

Threshold (Hz):

Virus ID Start (1–100): Virus ID End (1–100):

Run Comparison

Sensor Frequencies

Base Frequency (Hz): 1694137.6062395119

Measured Frequency (Hz): 1694138.316002553

They learn:

- How binding events shift frequency
- How biosensing features are generated
- How ML models use these features

9. Side-by-Side V2 & V6 Comparison Summary

Comparison Summary		
Feature	V2 (XBoost)	V6 (RandomForest)
Dataset size	12,000	120,000
Accuracy	88.5%	94.2%
Noise tolerance	Medium	High
Virus identification	Partial	Complete
Educational use	Intro ML	Advanced ML

Students learn that RandomForest provides more comprehensive identification and higher stability than XBoost.

10. Student Laboratory Workflow

Step 1 — Run Simulation

Students call GET /simulate to obtain:

- Base frequency
- Measured frequency
- Biosensing features

Step 2 — Run V2 Classification

Students call GET /classify/v2 to analyze:

- Noise sensitivity
- Early-stage ML behavior
- Partial virus identification

Step 3 — Run V6 Classification

Students call GET /classify/v6 to analyze:

- High-accuracy predictions
- Full virus identification
- Probability vs mass
- Probability vs virus ID

Step 4 — Compare V2 vs V6

Students learn:

- Why RandomForest outperforms XBoost
- How dataset size affects accuracy
- How noise influences classification
- How ML models evolve

Step 5 — Write Lab Report

Students summarize:

- Sensor frequencies
- Noise effects
- Virus predictions
- Model comparison
- Learning outcomes

11. Publication & Contact Information

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References

- *Swagger.pdf* — Full API endpoint documentation
- *Folder Structure of Selected ML Models.pdf* — Directory layout and file organization